

Feedback-Based Shape Control of Deformable Linear Objects via a Simplified Model in Robotic Manipulation

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Problem Statement

Deformable Linear Objects (DLOs) such as cables or rods are widely used in robotic manipulation, but are difficult to model and control.

Key difficulty: nonlinear continuum mechanics

- ▶ high-dimensional configuration space
- ▶ strong deformation nonlinearities

Existing modeling approaches:

- ▶ **Cosserat rod models:** accurate but computationally expensive
- ▶ **Mass-spring models:** efficient but low physical fidelity

Fundamental trade-off:

Accuracy $\leftrightarrow$ Computational efficiency

Open problem: How to combine physical accuracy with real-time performance for robotic control?

Cosserat Rod Model

State variables: The rod is described by the centerline position $p(s)$ and orientation $R(s)$, the strain measures $v(s)$ and $u(s)$, and the internal efforts $n(s)$ and $m(s)$.

Kinematics

$$\frac{dp}{ds} = Rv, \quad \frac{dR}{ds} = R\hat{u}$$

Static equilibrium

$$\frac{dn}{ds} = -f, \quad \frac{dm}{ds} = -\dot{p} \times n$$

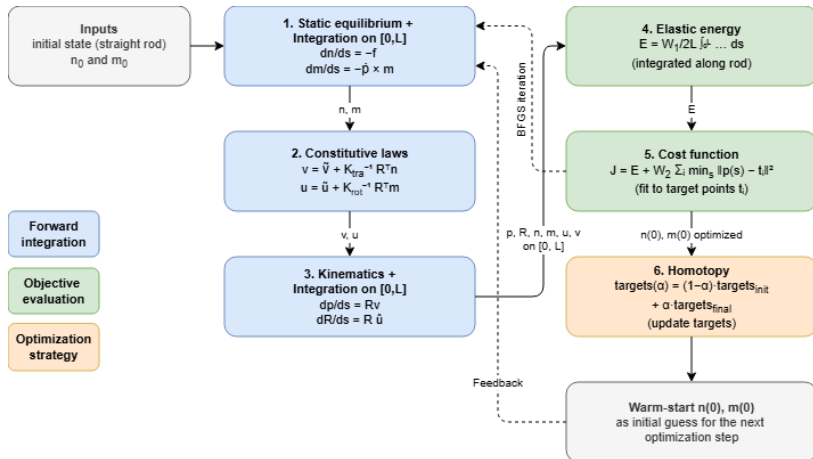
Constitutive laws:

$$n = K_{\text{tra}}(v - \tilde{v}), \quad m = K_{\text{rot}}(u - \tilde{u})$$

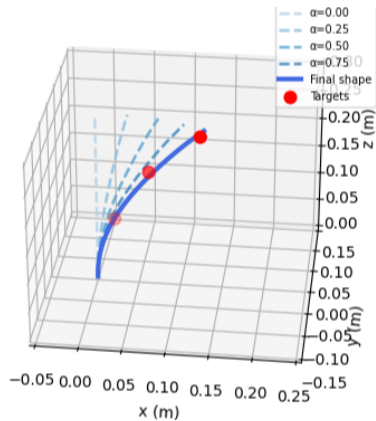
Elastic energy:

$$E = \frac{1}{2} \int_0^L \left[(v - \tilde{v})^T K_{\text{tra}}(v - \tilde{v}) + (u - \tilde{u})^T K_{\text{rot}}(u - \tilde{u}) \right] ds$$

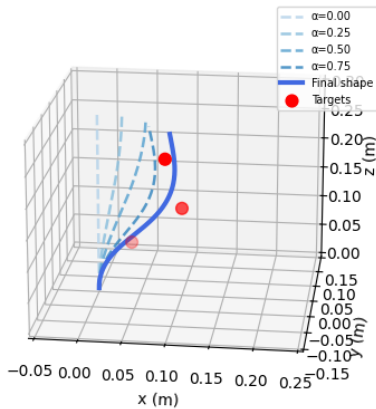
Shape Optimization



Results: Shape Optimization

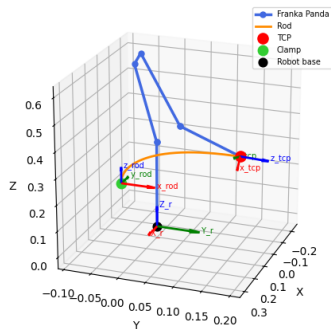
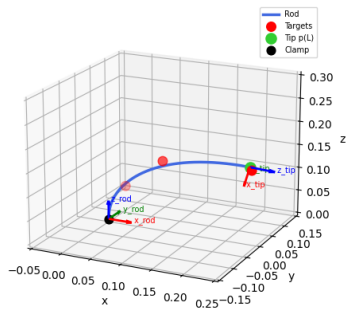


J-shape



C-shape

Results: Robotic Application



Mapping boundary conditions to robot end-effector

Proposed Approach

Objective: Improve simplified DLO models through model-based feedback correction using a high-fidelity reference.

Approach:

- 1. Feedforward prediction:** A simplified model (mass-spring) computes an initial actuation via optimization.
- 2. High-fidelity evaluation:** The resulting deformation is evaluated using a Cosserat rod model.
- 3. Feedback correction:** Shape errors are propagated back through a local actuation–shape mapping to update the control.

Outcome: Iterative refinement of actuation while maintaining computational efficiency.